Supplemental Materials for "Efficient Spin Seebeck and Spin Nernst Effects of Magnons in Altermagnets"

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I. SPIN-SPLIT DISPERSION OF ALTERMAGNETIC MAGNONS

The two-dimensional altermagnetic square lattice contains interpolated "a" and "b" sublattices, as illustrated in the main text. The spin Hamiltonian reads

$$\begin{aligned}\hat{H} = & J_1 \sum_{\langle i,j\rangle} \hat{\mathbf{S}}_{ai} \cdot \hat{\mathbf{S}}_{bj} \\ & + J_2 \sum_{\langle i_x,j_x\rangle} \hat{\mathbf{S}}_{ai} \cdot \hat{\mathbf{S}}_{aj} + J_2' \sum_{\langle i_y,j_y\rangle} \hat{\mathbf{S}}_{ai} \cdot \hat{\mathbf{S}}_{aj} + K \sum_i \left(\hat{S}_{ai}^z\right)^2 \\ & + J_2' \sum_{\langle i_x,j_x\rangle} \hat{\mathbf{S}}_{bi} \cdot \hat{\mathbf{S}}_{bj} + J_2 \sum_{\langle i_y,j_y\rangle} \hat{\mathbf{S}}_{bi} \cdot \hat{\mathbf{S}}_{bj} + K \sum_i \left(\hat{S}_{bi}^z\right)^2, \end{aligned} \tag{S1}$$

where the first term $J_1 > 0$ describes the antiferromagnetic spin interaction between"a" and "b" sublattices, and the second and third terms $\{J_2, J_2'\} < 0$ contain the ferromagnetic spin interactions within the sublattices. $\langle i,j\rangle$, $\langle i_x,j_x\rangle$, and $\langle i_y,j_y\rangle$ indicate the nearest-neighboring coupling between "a" and "b" sites along the *diagonal*, $\hat{\mathbf{x}}$, and $\hat{\mathbf{y}}$ direction, respectively. The anisotropy constant $K < 0$ favors out-of-plane magnetization. In terms of the bosonic operators $\hat{a}_i$ and $\hat{b}_i$ for the sublattices "a" and "b", the Holstein-Primakoff transformation [1] in the linear regime reads

$$\hat{S}_{ai}^+ \approx \sqrt{2S}\hat{a}_i, \quad \hat{S}_{ai}^- \approx \sqrt{2S}\hat{a}_i^\dagger, \quad \hat{S}_{ai}^z = S - \hat{a}_i^\dagger \hat{a}_i, \tag{S2a}$$

$$\hat{S}_{bj}^+ \approx \sqrt{2S}\hat{b}_j^\dagger, \quad \hat{S}_{bj}^- \approx \sqrt{2S}\hat{b}_j, \quad \hat{S}_{bj}^z = \hat{b}_j^\dagger \hat{b}_j - S, \tag{S2b}$$

where $\hat{S}_i^\pm = \hat{S}_i^x \pm i\hat{S}_i^y$. Disregarding the zero-point energy, the spin Hamiltonian (S1) becomes

$$\begin{aligned}\hat{H} = & J^* S \left(\sum_i \hat{a}_i^\dagger \hat{a}_i + \sum_j \hat{b}_j^\dagger \hat{b}_j\right) + 2J_1 S \sum_{\langle i,j\rangle} \left(\hat{a}_i \hat{b}_j + \hat{a}_i^\dagger \hat{b}_j^\dagger\right) \\ & + J_2 S \sum_{\langle i_x,j_x\rangle} \left(\hat{a}_i \hat{a}_j^\dagger + \hat{a}_i^\dagger \hat{a}_j\right) + J_2' S \sum_{\langle i_y,j_y\rangle} \left(\hat{a}_i \hat{a}_j^\dagger + \hat{a}_i^\dagger \hat{a}_j\right) \\ & + J_2' S \sum_{\langle i_x,j_x\rangle} \left(\hat{b}_i \hat{b}_j^\dagger + \hat{b}_i^\dagger \hat{b}_j\right) + J_2 S \sum_{\langle i_y,j_y\rangle} \left(\hat{b}_i \hat{b}_j^\dagger + \hat{b}_i^\dagger \hat{b}_j\right), \end{aligned} \tag{S3}$$

where the on-site energy $J^* = 4J_1 - 2J_2 - 2J_2' - 2K$. With $\hat{a}_i = \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{R}_i}\hat{a}_{\mathbf{k}}$ and $\hat{b}_j = \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{R}_j}\hat{b}_{\mathbf{k}}$, we transform to the wave-vector space:

$$\hat{H} = \sum_{\mathbf{k}} \left(\hat{a}_{\mathbf{k}}^\dagger, \hat{b}_{\mathbf{k}}\right) \hat{H}_{\mathbf{k}} \begin{pmatrix} \hat{a}_{\mathbf{k}} \\ \hat{b}_{\mathbf{k}}^\dagger \end{pmatrix}, \quad \hat{H}_{\mathbf{k}} = S \begin{pmatrix} J^* + 2J_2\gamma_1 + 2J_2'\gamma_2 & 4J_1\gamma_3 \\ 4J_1\gamma_3 & J^* + 2J_2'\gamma_1 + 2J_2\gamma_2 \end{pmatrix}, \tag{S4}$$

where $\gamma_1 = \cos(k_x a)$, $\gamma_2 = \cos(k_y a)$, and $\gamma_3 = \cos(k_x a/2)\cos(k_y a/2)$.

We find the magnon modes of Eq. (S4) by the Bogoliubov transformation

$$\begin{pmatrix} \hat{a}_{\mathbf{k}} \\ \hat{b}_{\mathbf{k}}^\dagger \end{pmatrix} = T_{\mathbf{k}} \begin{pmatrix} \hat{\alpha}_{\mathbf{k}} \\ \hat{\beta}_{\mathbf{k}}^\dagger \end{pmatrix}, \tag{S5}$$

where the paraunitary matrix

$$T_{\mathbf{k}} = \begin{pmatrix} u_{\mathbf{k}} & v_{\mathbf{k}} \\ v_{\mathbf{k}} & u_{\mathbf{k}} \end{pmatrix}, \tag{S6}$$

leading to $[J^* + (J_2 + J_2')(\gamma_1 + \gamma_2)]\, u_{\mathbf{k}} v_{\mathbf{k}} + 2J_1\gamma_3(u_{\mathbf{k}}^2 + v_{\mathbf{k}}^2) = 0$. On the other hand, the bosonic commutation relation for the magnon operators leads to the relation $T_{\mathbf{k}}^\dagger \hat{\sigma}_z T_{\mathbf{k}} = \hat{\sigma}_z$ with $\hat{\sigma}_z = \mathrm{diag}(1, -1)$ that requests $u_{\mathbf{k}}^2 - v_{\mathbf{k}}^2 = 1$. We thus obtain

$$u_{\mathbf{k}} = \sqrt{(\Delta^{-1} + 1)/2}, \quad v_{\mathbf{k}} = -\sqrt{(\Delta^{-1} - 1)/2}, \quad \Delta = \sqrt{1 - {\gamma_e}^2}, \tag{S7}$$

where the effective form factor $\gamma_e = 4J_1\gamma_3/[4J_1 - 2K - (J_2 + J_2')(2 - \gamma_1 - \gamma_2)]$. The magnon Hamiltonian contains the α- and β-modes with

$$\hat{H} = \sum_{\mathbf{k}} \left(\hat{\alpha}_{\mathbf{k}}^\dagger, \hat{\beta}_{\mathbf{k}}\right) \begin{pmatrix} \hbar\omega_{\mathbf{k}}^\alpha & 0 \\ 0 & \hbar\omega_{\mathbf{k}}^\beta \end{pmatrix} \begin{pmatrix} \hat{\alpha}_{\mathbf{k}} \\ \hat{\beta}_{\mathbf{k}}^\dagger \end{pmatrix}, \tag{S8}$$

where the eigenfrequency of α- and β-modes

$$\begin{aligned} \omega_{\mathbf{k}}^\alpha &= \frac{S}{\hbar}\left[(J_2 - J_2')(\gamma_1 - \gamma_2) + \frac{4J_1\gamma_3\Delta}{\gamma_e}\right], \\ \omega_{\mathbf{k}}^\beta &= \frac{S}{\hbar}\left[(J_2' - J_2)(\gamma_1 - \gamma_2) + \frac{4J_1\gamma_3\Delta}{\gamma_e}\right], \end{aligned} \tag{S9}$$

are not degenerate since $J_2 \neq J_2'$.

The α and β modes turn out to carry opposite "spins". Via the Holstein-Primakoff and Bogoliubov transformations, the z-component of the total spin operator reads

$$\begin{aligned} \hat{S}^z &\equiv \sum_i \left(\hat{S}_{ai}^z + \hat{S}_{bi}^z\right) = \sum_i \left(-\hat{a}_i^\dagger \hat{a}_i + \hat{b}_i^\dagger \hat{b}_i\right) = \sum_{\mathbf{k}} \left(-\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \hat{b}_{\mathbf{k}}^\dagger \hat{b}_{\mathbf{k}}\right) \\ &= \sum_{\mathbf{k}} \left(-\hat{\alpha}_{\mathbf{k}}^\dagger \hat{\alpha}_{\mathbf{k}} + \hat{\beta}_{\mathbf{k}}^\dagger \hat{\beta}_{\mathbf{k}}\right). \end{aligned} \tag{S10}$$

Consequently, the spin angular momentum (with the unit of $\hbar$) along the $\hat{\mathbf{z}}$-direction for any α and β modes

$$\langle 0|\hat{\alpha}_{\mathbf{k}} \hat{S}^z \hat{\alpha}_{\mathbf{k}}^\dagger|0\rangle = -1, \tag{S11a}$$

$$\langle 0|\hat{\beta}_{\mathbf{k}} \hat{S}^z \hat{\beta}_{\mathbf{k}}^\dagger|0\rangle = 1, \tag{S11b}$$

where $|0\rangle$ indicates the magnon vacuum.

II. DETAILS OF FIRST-PRINCIPLES CALCULATIONS

The first-principles calculations are implemented in the Vienna *ab initio* Simulation Package (VASP), based on the density functional theory (DFT) [2–4]. The exchange correlation functional is treated within the generalized gradient approximation (GGA) of Perdew-Burke-Ernzerhof form [5]. For describing the strong correlation effects of $3d$ electrons, the GGA+U method is applied with an effective Hubbard U parameter that is chosen to be 4 eV for transition mental elements. U_{eff}=3, 5 eV are also tested, and the main findings about the material candidate are robust against U_{eff} values. The crystal structure is fully relaxed until the Hellmann-Feynman force acting on each atom is less than 10^{-3} eV/Å. The plane-wave cutoff energy is set to be 420 eV and the two-dimensional Brillouin zone is sampled using a Γ-centered 20×20×1 Monkhorst-pack k mesh. In order to avoid interactions between adjacent layers, the vacuum space is set to be larger than 15 Å. The finite displacement method is applied for obtaining the phonon band dispersion in a 4×4×1 supercell,

based on PHONOPY code [6]. *Ab initio* molecular dynamics simulations are performed in the canonical NVT particles ensemble used with Nosé thermostat [7]. Such simulations are conducted for 10 ps at 500 K. The energy mapping methods are applied for extracting magnetic parameters [8, 9]. Specifically, the self-consistent energies of a 2×2×1 supercell with spin configurations FM, AFM1, AFM2, and AFM3 (Fig. S1) are calculated, and accordingly, the nearest-neighboring exchange coupling J_1 and next-nearest-neighboring exchange coupling $J_2(J_2')$ reads $J_1 = (E_{FM} - E_{AFM1})/32S^2$, $J_2 = (E_{AFM1} + E_{FM} - 2E_{AFM2})/32S^2$ and $J_2' = (E_{AFM1} + E_{FM} - 2E_{AFM3})/32S^2$. The magnetic anisotropy is estimated by the energy difference of the unit cell with out-of-plane magnetization along $\hat{\mathbf{z}}$ and in-plane magnetization along $\hat{\mathbf{x}}$. Cr_2Se_2O possesses the same structure with Cr_2Te_2O, and the optimizted lattice constant $a = 4.04$ Å. The bonding angle of Cr–Se–Cr and Cr-O-Cr equal 100° and 180°, respectively.

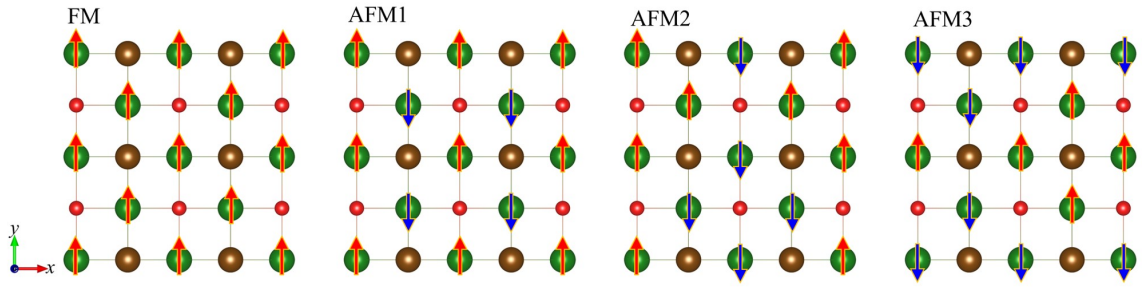


FIG. S1. Spin configurations, indicated by red and blue arrows, for FM, AFM1, AFM2, and AFM3 that are chosen for extracting exchange couplings between Cr atoms in Cr_2Te_2O and Cr_2Se_2O monolayers.

III. KUBO THEORY FOR SPIN AND THERMAL CONDUCTIVITIES

The magnon thermal conductivity tensor is defined via the relation

$$\left\langle \hat{j}_m^z \right\rangle = \sigma_{mn}(-\partial_n T), \tag{S12}$$

where $\left\langle \hat{j}_m^z \right\rangle$ represents the spin current density flowing along the $\hat{\mathbf{m}}$-direction and $\partial_n T$ represents the temperature gradient along n-direction. In terms of Kubo formula [10, 11],

$$\left\langle \hat{j}_m^z \right\rangle = \lim_{\Omega\to 0} \left\{ \left[\Pi_{mn}^R(\Omega) - \Pi_{mn}^R(0) \right] / \mathrm{i}\Omega \right\} \frac{-\hbar \partial_n T}{AT}, \tag{S13}$$

where $\Pi_{mn}^R(\Omega)$ is the retarded current-current correlation function, Ω is the frequency, and A represents the sample area. Substituting (S12) into (S13) yields

$$\sigma_{mn} = \frac{\hbar}{AT} \lim_{\Omega\to 0} \left\{ \left[\Pi_{mn}^R(\Omega) - \Pi_{mn}^R(0) \right] / \mathrm{i}\Omega \right\}. \tag{S14}$$

$\Pi_{mn}^R(\Omega)$ is further evaluated in the Matsubara representation [11]:

$$\begin{aligned} \Pi_{mn}^R(\Omega) &= \Pi_{mn}(\mathrm{i}\Omega \to \Omega + \mathrm{i}0^+), \\ \Pi_{mn}(\mathrm{i}\Omega) &= -\int_0^\beta \mathrm{e}^{\mathrm{i}\tau\Omega} \left\langle \hat{\mathcal{T}} \hat{J}_m(\tau) \hat{Q}_n(0) \right\rangle \mathrm{d}\tau, \end{aligned} \tag{S15}$$

where $\beta = 1/(k_B T)$, τ is the imaginary time, $\hat{\mathcal{T}}$ is the time-ordering operator, and $\Omega = 2\pi l/\beta$ is the bosonic Matsubara frequency with integer l. In terms of magnon basis, the total spin and thermal current reads [12–14]

$$\hat{J}_m = \sum_{\mathbf{k}} \hat{\Psi}^\dagger_{\mathbf{k}} \hat{\mathcal{J}}_{m,\mathbf{k}} \hat{\Psi}_{\mathbf{k}}, \quad \hat{\mathcal{J}}_{m,\mathbf{k}} = \frac{1}{2} T^\dagger_{\mathbf{k}} \left(\hat{v}_{m,\mathbf{k}} \hat{\sigma}_z \hat{S}^z + \hat{S}^z \hat{\sigma}_z \hat{v}_{m,\mathbf{k}} \right) T_{\mathbf{k}} = -T^\dagger_{\mathbf{k}} \left(\partial_{k_m} \hat{H}_{\mathbf{k}} \right) T_{\mathbf{k}}, \tag{S16a}$$

$$\hat{Q}_n = \sum_{\mathbf{k}} \hat{\Psi}^\dagger_{\mathbf{k}} \hat{\mathcal{Q}}_{n,\mathbf{k}} \hat{\Psi}_{\mathbf{k}}, \quad \hat{\mathcal{Q}}_{n,\mathbf{k}} = \frac{1}{2} T^\dagger_{\mathbf{k}} \left(\hat{v}_{n,\mathbf{k}} \hat{\sigma}_z \hat{H}_{\mathbf{k}} + \hat{H}_{\mathbf{k}} \hat{\sigma}_z \hat{v}_{n,\mathbf{k}} \right) T_{\mathbf{k}}, \tag{S16b}$$

where velocity operator $\hat{v}_{m,\mathbf{k}} = (\partial_{k_m} \hat{H}_{\mathbf{k}})/\hbar$. Following Ref. [13], we transform (S15) to wave-vector space:

$$\begin{aligned}
\Pi_{mn}(\mathrm{i}\Omega) &= -\int_0^\beta \mathrm{e}^{\mathrm{i}\tau\Omega} \sum_{\mathbf{k},\mathbf{k}'} \sum_{a,b,c,d} (\hat{\mathcal{J}}_{m,\mathbf{k}})_{ab} (\hat{\mathcal{Q}}_{n,\mathbf{k}'})_{cd} \left\langle \hat{\mathcal{T}} \hat{\Psi}^\dagger_{a,\mathbf{k}}(\tau) \hat{\Psi}_{b,\mathbf{k}}(\tau) \hat{\Psi}^\dagger_{c,\mathbf{k}'}(0) \hat{\Psi}_{d,\mathbf{k}'}(0) \right\rangle \mathrm{d}\tau \\
&= -\int_0^\beta \mathrm{e}^{\mathrm{i}\tau\Omega} \sum_{\mathbf{k},\mathbf{k}'} \sum_{a,b,c,d} (\hat{\mathcal{J}}_{m,\mathbf{k}})_{ab} (\hat{\mathcal{Q}}_{n,\mathbf{k}'})_{cd} \\
&\left(\left\langle \hat{\mathcal{T}} \hat{\Psi}^\dagger_{a,\mathbf{k}}(\tau) \hat{\Psi}^\dagger_{c,\mathbf{k}'}(0) \right\rangle \left\langle \hat{\mathcal{T}} \hat{\Psi}_{b,\mathbf{k}}(\tau) \hat{\Psi}_{d,\mathbf{k}'}(0) \right\rangle + \left\langle \hat{\mathcal{T}} \hat{\Psi}^\dagger_{a,\mathbf{k}}(\tau) \hat{\Psi}_{d,\mathbf{k}'}(0) \right\rangle \left\langle \hat{\mathcal{T}} \hat{\Psi}_{b,\mathbf{k}}(\tau) \hat{\Psi}^\dagger_{c,\mathbf{k}'}(0) \right\rangle \right) \mathrm{d}\tau,
\end{aligned} \tag{S17}$$

where $a, b, c, d \in \{\alpha, \beta\}$ indicates the band index. For the bosonic operators with diagonal basis, the time-ordering ($\tau > 0$) average of equilibrium states follows

$$\left\langle \hat{\mathcal{T}} \hat{\alpha}^\dagger_{\mathbf{k}}(\tau' + \tau) \hat{\beta}_{\mathbf{k}'}(\tau') \right\rangle = \left\langle \hat{\mathcal{T}} \hat{\alpha}_{\mathbf{k}}(\tau' + \tau) \hat{\beta}^\dagger_{\mathbf{k}'}(\tau') \right\rangle = 0, \tag{S18a}$$

$$\begin{aligned}
\left\langle \hat{\mathcal{T}} \hat{\alpha}^\dagger_{\mathbf{k}}(\tau' + \tau) \hat{\alpha}_{\mathbf{k}'}(\tau') \right\rangle &= \delta_{\mathbf{k},\mathbf{k}'} g\left(\hbar\omega^\alpha_{\mathbf{k}}\right) e^{\tau\hbar\omega^\alpha_{\mathbf{k}}}, \\
\left\langle \hat{\mathcal{T}} \hat{\alpha}_{\mathbf{k}}(\tau' + \tau) \hat{\alpha}^\dagger_{\mathbf{k}'}(\tau') \right\rangle &= -\delta_{\mathbf{k},\mathbf{k}'} g\left(-\hbar\omega^\alpha_{\mathbf{k}}\right) e^{-\tau\hbar\omega^\alpha_{\mathbf{k}}},
\end{aligned} \tag{S18b}$$

with Bose-Einstein distribution $g\left(\hbar\omega^\alpha_{\mathbf{k}}\right) = (e^{\hbar\omega^\alpha_{\mathbf{k}}/k_B T} - 1)^{-1}$. Accordingly,

$$\begin{aligned}
\Pi_{mn}(\mathrm{i}\Omega) &= -\int_0^\beta \mathrm{e}^{\mathrm{i}\tau\Omega} \sum_{\mathbf{k},\mathbf{k}'} \sum_{a,b,c,d} \left(\hat{\mathcal{J}}_{m,\mathbf{k}}\right)_{ab} \left(\hat{\mathcal{Q}}_{n,\mathbf{k}'}\right)_{cd} \left\langle \hat{\mathcal{T}} \hat{\Psi}^\dagger_{a,\mathbf{k}}(\tau) \hat{\Psi}_{d,\mathbf{k}'}(0) \right\rangle \left\langle \hat{\mathcal{T}} \hat{\Psi}_{b,\mathbf{k}}(\tau) \hat{\Psi}^\dagger_{c,\mathbf{k}'}(0) \right\rangle \mathrm{d}\tau \\
&= \sum_{\mathbf{k}} \sum_{a,b} \left(\hat{\mathcal{J}}_{m,\mathbf{k}}\right)_{ab} \left(\hat{\mathcal{Q}}_{n,\mathbf{k}}\right)_{ba} (\hat{\sigma}_z)_{aa} (\hat{\sigma}_z)_{bb} \frac{g\left[(\hat{\sigma}_z \mathcal{E}_{\mathbf{k}})_{bb}\right] - g\left[(\hat{\sigma}_z \mathcal{E}_{\boldsymbol{k}})_{aa}\right]}{(\hat{\sigma}_z \mathcal{E}_{\mathbf{k}})_{aa} - (\hat{\sigma}_z \mathcal{E}_{\mathbf{k}})_{bb} + \mathrm{i}\Omega},
\end{aligned} \tag{S19}$$

where eigenmatrix

$$\mathcal{E}_{\mathbf{k}} = \begin{pmatrix} \hbar\omega^\alpha_{\mathbf{k}} & 0 \\ 0 & \hbar\omega^\beta_{\mathbf{k}} \end{pmatrix}. \tag{S20}$$

We substitute (S19) into (S14) with $i0^+ \to i\Gamma$, where Γ governs quasiparticle broadening, and obtain the magnon thermal conductivity

$$\sigma_{mn} = -\frac{\hbar}{AT} \sum_{\mathbf{k}} \sum_{a,b} \mathrm{Re}\left[\left(\hat{\mathcal{J}}_{m,\mathbf{k}}\right)_{ab} \left(\hat{\mathcal{Q}}_{n,\mathbf{k}}\right)_{ba} \right] \frac{g\left[(\hat{\sigma}_z \mathcal{E}_{\mathbf{k}})_{aa}\right] - g\left[(\hat{\sigma}_z \mathcal{E}_{\mathbf{k}})_{bb}\right]}{(\hat{\sigma}_z \mathcal{E}_{\mathbf{k}})_{aa} - (\hat{\sigma}_z \mathcal{E}_{\mathbf{k}})_{bb}} \frac{\Gamma\,(\hat{\sigma}_z)_{aa}\,(\hat{\sigma}_z)_{bb}}{\Gamma^2 + \left[(\hat{\sigma}_z \mathcal{E}_{\mathbf{k}})_{aa} - (\hat{\sigma}_z \mathcal{E}_{\mathbf{k}})_{bb}\right]^2}. \tag{S21}$$

By defining magnon lifetime $\tau_0 = \hbar/\Gamma$ and disregarding magnon-magnon interaction at low temperature [14, 15], we arrive at

$$\begin{aligned}
\sigma_{mn} = &-\frac{\tau_0}{A k_B T^2} \sum_{\mathbf{k}} \left(T^\dagger_{\mathbf{k}} (\partial_{k_m} \hat{H}_{\mathbf{k}}) T_{\mathbf{k}}\right)_{\alpha\alpha} \left(T^\dagger_{\mathbf{k}} (\partial_{k_n} \hat{H}_{\mathbf{k}}) T_{\mathbf{k}}\right)_{\alpha\alpha} \omega^\alpha_{\mathbf{k}} \frac{e^{\hbar\omega^\alpha_{\mathbf{k}}/k_B T}}{\left(e^{\hbar\omega^\alpha_{\mathbf{k}}/k_B T} - 1\right)^2} \\
&+ \frac{\tau_0}{A k_B T^2} \sum_{\mathbf{k}} \left(T^\dagger_{\mathbf{k}} (\partial_{k_m} \hat{H}_{\mathbf{k}}) T_{\mathbf{k}}\right)_{\beta\beta} \left(T^\dagger_{\mathbf{k}} (\partial_{k_n} \hat{H}_{\mathbf{k}}) T_{\mathbf{k}}\right)_{\beta\beta} \omega^\beta_{\mathbf{k}} \frac{e^{-\hbar\omega^\beta_{\mathbf{k}}/k_B T}}{\left(e^{-\hbar\omega^\beta_{\mathbf{k}}/k_B T} - 1\right)^2}.
\end{aligned} \tag{S22}$$

A similar procedure is applied for the calculation of thermal conductivity κ_{mn}. Different from spin conductivity, here $\Pi_{mn}(\mathrm{i}\Omega)$ is rewritten as

$$\Pi_{mn}(\mathrm{i}\Omega) = -\int_0^\beta \mathrm{e}^{\mathrm{i}\tau\Omega}\left\langle \hat{\mathcal{T}}\hat{Q}_m(\tau)\hat{Q}_n(0)\right\rangle \mathrm{d}\tau. \tag{S23}$$

Transforming (S23) to wave-vector space,

$$\Pi_{mn}(\mathrm{i}\Omega) = \sum_{\mathbf{k}}\sum_{a,b}\left(\hat{\mathcal{Q}}_{m,\mathbf{k}}\right)_{ab}\left(\hat{\mathcal{Q}}_{n,\mathbf{k}}\right)_{ba}(\hat{\sigma}_z)_{aa}(\hat{\sigma}_z)_{bb}\frac{g\left[(\hat{\sigma}_z\mathcal{E}_{\mathbf{k}})_{bb}\right]-g\left[(\hat{\sigma}_z\mathcal{E}_{\boldsymbol{k}})_{aa}\right]}{(\hat{\sigma}_z\mathcal{E}_{\mathbf{k}})_{aa}-(\hat{\sigma}_z\mathcal{E}_{\mathbf{k}})_{bb}+\mathrm{i}\Omega}. \tag{S24}$$

Substituting (S24) into (S14), we obtain

$$\begin{aligned}\kappa_{mn} =& \frac{\tau_0}{Ak_BT^2}\sum_{\mathbf{k}}\left(T_{\mathbf{k}}^\dagger(\partial_{k_m}\hat{H}_{\mathbf{k}})T_{\mathbf{k}}\right)_{\alpha\alpha}\left(T_{\mathbf{k}}^\dagger(\partial_{k_n}\hat{H}_{\mathbf{k}})T_{\mathbf{k}}\right)_{\alpha\alpha}(\omega_{\mathbf{k}}^\alpha)^2\frac{e^{\hbar\omega_{\mathbf{k}}^\alpha/k_BT}}{\left(e^{\hbar\omega_{\mathbf{k}}^\alpha/k_BT}-1\right)^2}\\ &+\frac{\tau_0}{Ak_BT^2}\sum_{\mathbf{k}}\left(T_{\mathbf{k}}^\dagger(\partial_{k_m}\hat{H}_{\mathbf{k}})T_{\mathbf{k}}\right)_{\beta\beta}\left(T_{\mathbf{k}}^\dagger(\partial_{k_n}\hat{H}_{\mathbf{k}})T_{\mathbf{k}}\right)_{\beta\beta}\left(\omega_{\mathbf{k}}^\beta\right)^2\frac{e^{-\hbar\omega_{\mathbf{k}}^\beta/k_BT}}{\left(e^{-\hbar\omega_{\mathbf{k}}^\beta/k_BT}-1\right)^2}.\end{aligned} \tag{S25}$$

IV. BOLTZMANN TRANSPORT THEORY FOR SPIN AND THERMAL CONDUCTIVITIES

With the temperature gradient along the $\hat{\mathbf{n}}$ direction ∂_nT, the Boltzmann equation in the relaxation time τ_0 approximation is given by

$$\frac{g_{neq}-g_{eq}}{\tau_0} = -v_n\partial_nT\frac{\partial g_{eq}}{\partial T} = -v_n\partial_nT\frac{E}{k_BT^2}\frac{e^{E/k_BT}}{\left(e^{E/k_BT}-1\right)^2}, \tag{S26}$$

where the equilibrium thermal distribution $g_{eq} = (e^{E/k_BT}-1)^{-1}$. The consequent spin-up current density along the $\hat{\mathbf{m}}$ direction reads

$$\left\langle\hat{j}_m^z\right\rangle = \frac{v_m\hbar}{A}(g_{neq}-g_{eq}) = -\frac{\tau_0E}{Ak_BT^2}\hbar v_mv_n\partial_nT\frac{e^{E/k_BT}}{\left(e^{E/k_BT}-1\right)^2}. \tag{S27}$$

Accordingly,

$$\sigma_{mn} = \sigma_{mn}^\alpha+\sigma_{mn}^\beta, \tag{S28a}$$

$$\sigma_{mn}^\alpha = -\frac{\tau_0}{Ak_BT^2}\sum_{\mathbf{k}}\left(T_{\mathbf{k}}^\dagger(\partial_{k_m}\hat{H}_{\mathbf{k}})T_{\mathbf{k}}\right)_{\alpha\alpha}\left(T_{\mathbf{k}}^\dagger(\partial_{k_n}\hat{H}_{\mathbf{k}})T_{\mathbf{k}}\right)_{\alpha\alpha}\omega_{\mathbf{k}}^\alpha\frac{e^{\hbar\omega_{\mathbf{k}}^\alpha/k_BT}}{\left(e^{\hbar\omega_{\mathbf{k}}^\alpha/k_BT}-1\right)^2}, \tag{S28b}$$

$$\sigma_{mn}^\beta = \frac{\tau_0}{Ak_BT^2}\sum_{\mathbf{k}}\left(T_{\mathbf{k}}^\dagger(\partial_{k_m}\hat{H}_{\mathbf{k}})T_{\mathbf{k}}\right)_{\beta\beta}\left(T_{\mathbf{k}}^\dagger(\partial_{k_n}\hat{H}_{\mathbf{k}})T_{\mathbf{k}}\right)_{\beta\beta}\omega_{\mathbf{k}}^\beta\frac{e^{\hbar\omega_{\mathbf{k}}^\beta/k_BT}}{\left(e^{\hbar\omega_{\mathbf{k}}^\beta/k_BT}-1\right)^2}. \tag{S28c}$$

With the identity

$$\frac{e^{E/k_BT}}{\left(e^{E/k_BT}-1\right)^2} = \frac{e^{-E/k_BT}}{\left(e^{-E/k_BT}-1\right)^2}, \tag{S29}$$

we find the same spin conductivity derived based on the Kubo formula by constant lifetime approximation and restricting to intraband contribution [see Eq. S22].

The same approach is adopted for resolving the thermal conductivity by defining the thermal current density

$$\left\langle\hat{j}_m^t\right\rangle = \frac{v_mE}{A}(g_{neq}-g_{eq}) = -\frac{\tau_0E^2}{Ak_BT^2}v_mv_n\partial_nT\frac{e^{E/k_BT}}{\left(e^{E/k_BT}-1\right)^2}. \tag{S30}$$

Accordingly,

$$\begin{aligned}
\kappa_{mn} &= \kappa^{\alpha}_{mn} + \kappa^{\beta}_{mn}, \\
\kappa^{\alpha}_{mn} &= \frac{\tau_0}{Ak_BT^2}\sum_{\mathbf{k}}\left(T^{\dagger}_{\mathbf{k}}(\partial_{k_m}\hat{H}_{\mathbf{k}})T_{\mathbf{k}}\right)_{\alpha\alpha}\left(T^{\dagger}_{\mathbf{k}}(\partial_{k_n}\hat{H}_{\mathbf{k}})T_{\mathbf{k}}\right)_{\alpha\alpha}(\omega^{\alpha}_{\mathbf{k}})^2\frac{e^{\hbar\omega^{\alpha}_{\mathbf{k}}/k_BT}}{\left(e^{\hbar\omega^{\alpha}_{\mathbf{k}}/k_BT}-1\right)^2}, \\
\kappa^{\beta}_{mn} &= \frac{\tau_0}{Ak_BT^2}\sum_{\mathbf{k}}\left(T^{\dagger}_{\mathbf{k}}(\partial_{k_m}\hat{H}_{\mathbf{k}})T_{\mathbf{k}}\right)_{\beta\beta}\left(T^{\dagger}_{\mathbf{k}}(\partial_{k_n}\hat{H}_{\mathbf{k}})T_{\mathbf{k}}\right)_{\beta\beta}\left(\omega^{\beta}_{\mathbf{k}}\right)^2\frac{e^{\hbar\omega^{\beta}_{\mathbf{k}}/k_BT}}{\left(e^{\hbar\omega^{\beta}_{\mathbf{k}}/k_BT}-1\right)^2}.
\end{aligned} \tag{S31}$$

V. PROPERTIES OF ALTERMAGNETS AND ALTERMAGNETIC MAGNONS

Figure S2 shows the magnetic and structural properties of Cr_2Te_2O and Cr_2Se_2O. The group velocity of altermagnetic magnons is shown in Fig. S3. We plot the magnon band dispersion of Cr_2Se_2O in Fig. S4.

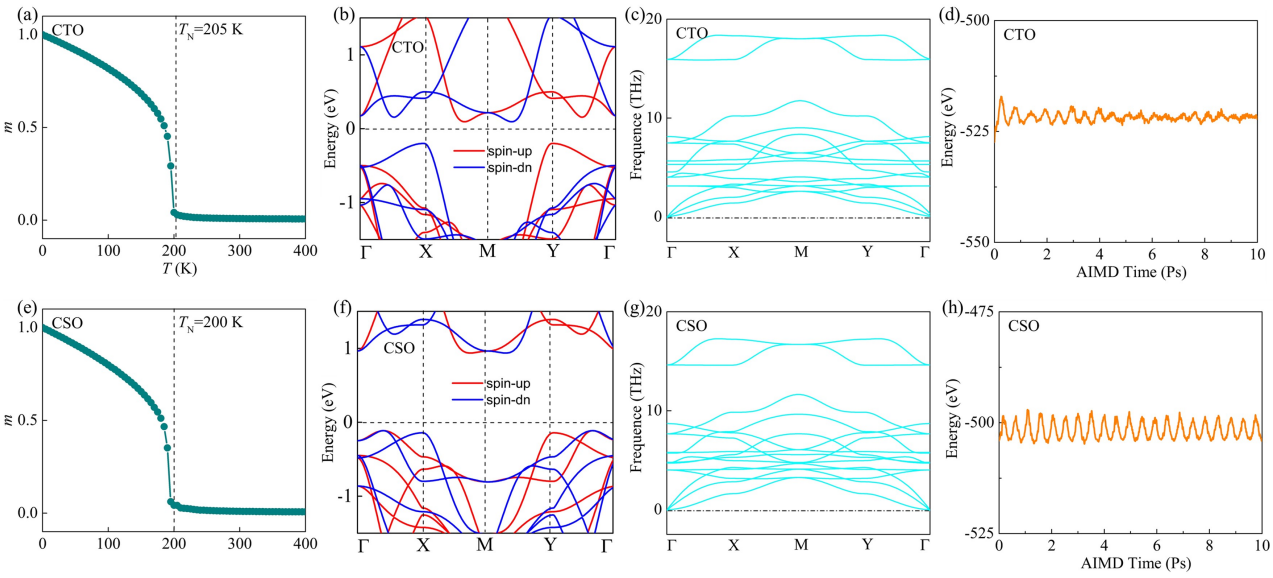


FIG. S2. (a) and (c) Monte Carlo simulations of temperature-dependent Néel magnetization. Net magnetization of the antiferromagnetic system equals zero indicating the atomic spin of each sublattice is aligned. Accordingly, the normalized magnetization m_1 and m_2 of two sublattices are defined and m=(m_1+m_2)/2 is applied to estimate the Néel temperature. (b) and (f) Spin-polarized band structure. The indirect band gap of Cr_2Te_2O and Cr_2Se_2O reach 0.30 and 1.05 eV, respectively. (c) and (g) Phonon band dispersions. (d) and (h) Evolution of total energy with temperature $T = 500$ K during *ab initio* molecular dynamics simulations.

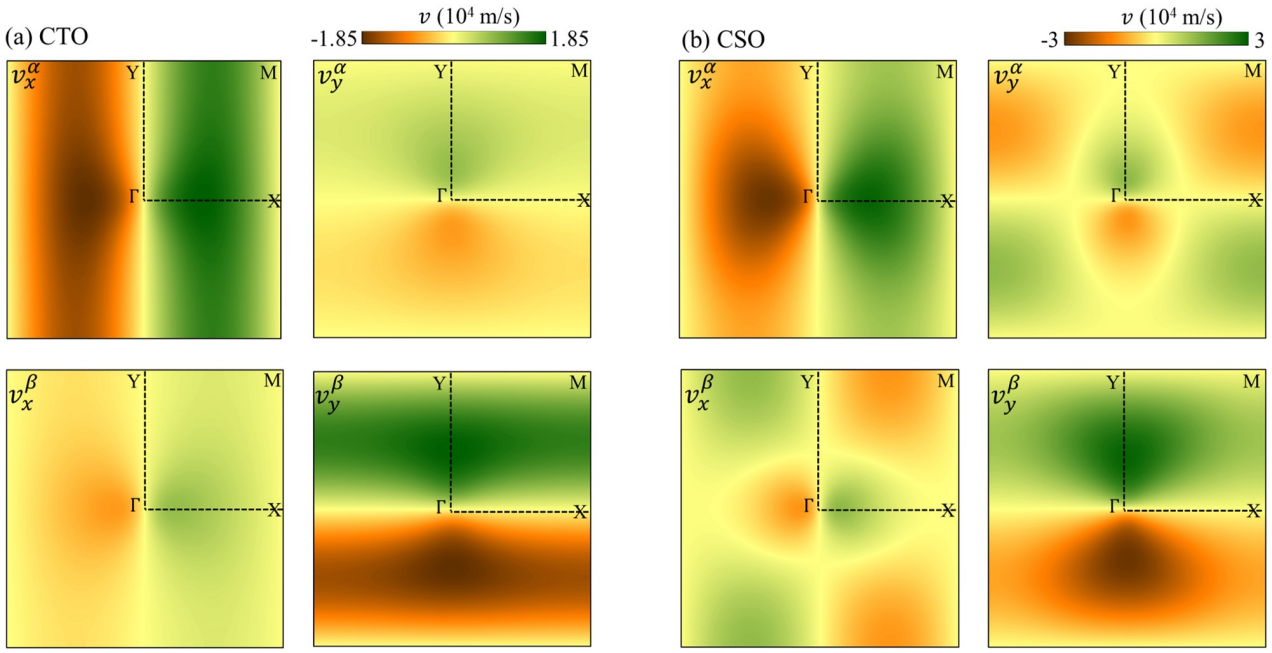


FIG. S3. The group velocity v of magnons of (a) Cr_2Te_2O and (b) Cr_2Se_2O. Specifically, $v_n^{\alpha} = (1/\hbar)(T_{\mathbf{k}}^{\dagger}(\partial_{k_n}\hat{H}_{\mathbf{k}})T_{\mathbf{k}})_{\alpha\alpha}$ and $v_n^{\beta} = (1/\hbar)(T_{\mathbf{k}}^{\dagger}(\partial_{k_n}\hat{H}_{\mathbf{k}})T_{\mathbf{k}})_{\beta\beta}$. Notably, the magnitude of v reaches 10^4 m/s. v exhibits significant anisotropy in $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ directions.

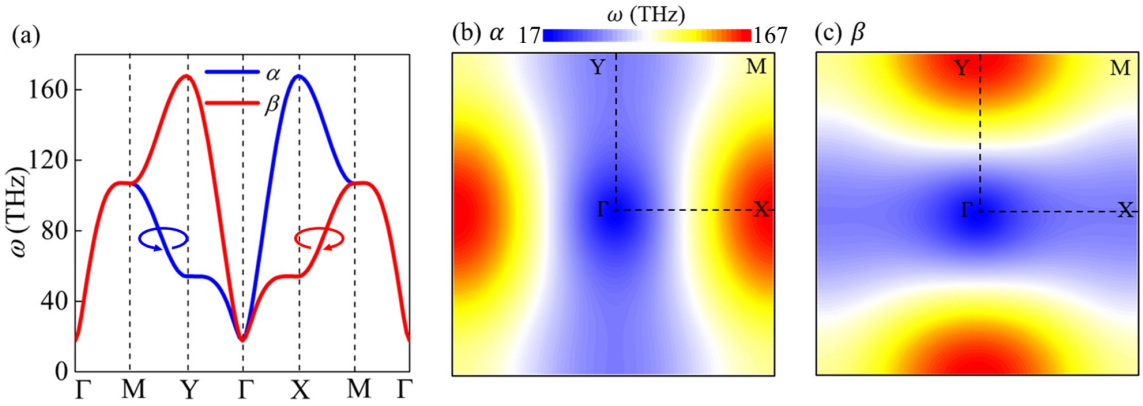


FIG. S4. (a) Spin-splitting magnon band structure and (b, c) magnon energy spectrum of Cr_2Se_2O. Similar to Cr_2Te_2O, anisotropic spin-momentum locking emerges, leading to the spin Seebeck and Nernst effects without the assistance of magnetic field and spin-orbit coupling. For $T = 100$ K and $\tau_0 = 10$ ps, the maximum magnitude of $\sigma_{\parallel(\perp)}$ reaches 1.62 meV/K.

VI. OTHER POTENTIAL CANDIDATES

Besides Cr_2Te_2O and Cr_2Se_2O, we find that Cr_2I_2O and Cr_2Br_2O [see crystal structure in Fig. S5(a)] also combines non-relativistic spin splitting and insulating features in electronic band structures [Fig. S5(b) and (c)]. By using energy mapping methods, we obtain $J_1 = 12.25$, $J_2 = -1.91$, $J_2' = 6.20$ and $K = -0.07$ meV for Cr_2I_2O, and $J_1 = 12.03$, $J_2 = -2.35$, $J_2' = 6.31$ and $K = 0.02$ meV for Cr_2Br_2O. Despite the anisotropic

nearest-neighboring exchange coupling that exists in both candidates, only Cr_2I_2O preserves the out-of-plane magnetization. The anisotropic spin-momentum locking [Fig. S5(d-f)] leads to the spin Seebeck and spin Nernst effects independent of the external magnetic field and spin-orbit coupling. For $T = 100$ K and $\tau_0 = 10$ ps, the maximum magnitude of $\sigma_{\parallel(\perp)}$ reaches1.64 meV/K.

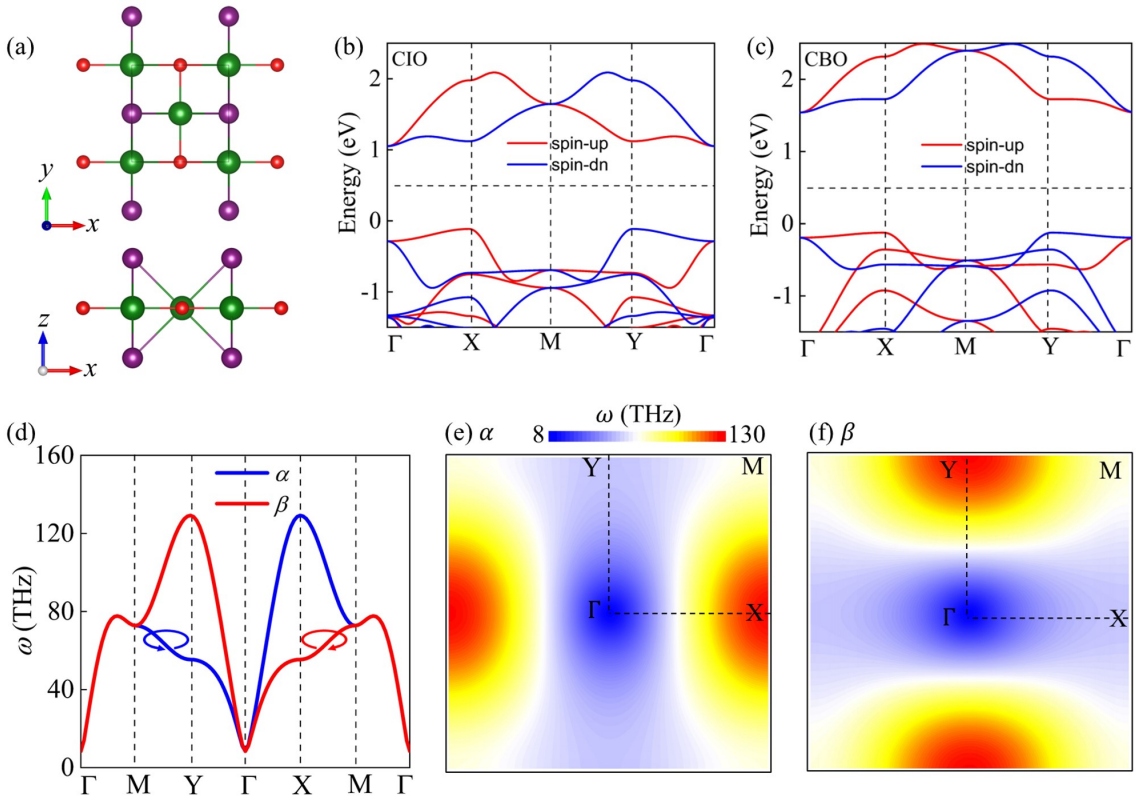


FIG. S5. (a) The top and side views of the crystal structure of Cr_2I_2O. The green, purple, and red balls represent the Cr, I, and O elements, respectively. (b) and (c) The spin-polarized band structures. The band gap of Cr_2I_2O and Cr_2Br_2O reaches 1.24 and 1.85 eV, respectively. (d) Spin-splitting magnon band structure and (e, f) magnon energy spectra of Cr_2I_2O.

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